

Ejercicios: SQL $\Leftrightarrow$ Álgebra Relacional

40 ejercicios sobre 4 mini-esquemas (Sailors–Boats–Reserves, Suppliers–Parts–Catalog, Employees–Supervises, Clients–Staff–Appointments).

Esquemas de trabajo

```
A - Sailors-Boats-Reserves (Ramakrishnan/Gehrke)
sailors(sid, sname, rating, age)
boats(bid, bname, color)
reserves(sid, bid, rdate)

B - Suppliers-Parts-Catalog (York, Bolzano)
suppliers(sid, sname, address)
parts(pid, pname, color)
catalog(sid, pid, cost)

C - Employees-Supervises (Toronto CSC343)
employees(number, name, age, salary)
supervises(boss, employee)

D - Clients-Staff-Appointments (Toronto CSC343)
clients(cid, name, phone)
staff(sid, name)
appointments(cid, adate, atime, service, sid)

FKs evidentes: reserves.sid -> sailors.sid, reserves.bid -> boats.bid,
catalog.sid -> suppliers.sid, catalog.pid -> parts.pid,
supervises.boss/employee -> employees.number,
appointments.cid -> clients.cid, appointments.sid -> staff.sid
```

Instrucciones

Sección 1: para cada consulta SQL, escribe la expresión equivalente en álgebra relacional ($\sigma, \pi, \bowtie, \cup, \cap, -, \times, \rho, \div$). Sección 2: para cada expresión de álgebra relacional, escribe el SQL equivalente. Verifica cada consulta ejecutándola contra tu instancia.

1 De SQL a Álgebra Relacional

1. Colores de los botes reservados por Albert.

```
SELECT DISTINCT b.color
FROM sailors s
JOIN reserves r ON r.sid = s.sid
JOIN boats b ON b.bid = r.bid
WHERE s.sname = 'Albert';
```

Respuesta:

$$\pi_{color}(\sigma_{sname='Albert'}(sailors) \bowtie reserves \bowtie boats)$$

2. sid de marineros con rating ≥ 8 o que reservaron el bote 103.

```
SELECT sid FROM sailors WHERE rating >= 8
UNION
SELECT sid FROM reserves WHERE bid = 103;
```

Respuesta:

$$\pi_{sid}(\sigma_{rating \geq 8}(sailors)) \cup \pi_{sid}(\sigma_{bid=103}(reserves))$$

3. Nombres de marineros que no reservaron un bote rojo.

```
SELECT s.sname
FROM sailors s
WHERE s.sid NOT IN (
    SELECT r.sid FROM reserves r JOIN boats b ON b.bid = r.bid
    WHERE b.color = 'red'
);
```

Respuesta:

$$\pi_{sname}(sailors) - \pi_{sname}(\sigma_{color='red'}(boats) \bowtie reserves \bowtie sailors)$$

4. sid de marineros con edad > 20 que no reservaron un bote rojo.

```
SELECT s.sid
FROM sailors s
WHERE s.age > 20
AND s.sid NOT IN (
    SELECT r.sid FROM reserves r JOIN boats b ON b.bid = r.bid
    WHERE b.color = 'red'
);
```

Respuesta:

$$\pi_{sid}(\sigma_{age > 20}(sailors)) - \pi_{sid}(\sigma_{color='red'}(boats) \bowtie reserves)$$

5. Nombres de marineros que reservaron al menos dos botes distintos.

```
SELECT DISTINCT s.sname
FROM sailors s
JOIN reserves r1 ON r1.sid = s.sid
JOIN reserves r2 ON r2.sid = s.sid
WHERE r1.bid <> r2.bid;
```

Respuesta:

$$\pi_{sname} (sailors \bowtie \pi_{sid} (\rho_{R1} (reserves) \bowtie_{R1.sid=R2.sid \wedge R1.bid \neq R2.bid} \rho_{R2} (reserves)))$$

6. Nombres de marineros que reservaron todos los botes.

```
SELECT s.sname
FROM sailors s
WHERE NOT EXISTS (
  SELECT 1 FROM boats b
  WHERE NOT EXISTS (
    SELECT 1 FROM reserves r WHERE r.sid = s.sid AND r.bid = b.bid
  )
);
```

Respuesta:

$$\pi_{sname} ((\pi_{sid,bid} (reserves) \div \pi_{bid} (boats)) \bowtie sailors)$$

7. Nombres de marineros que reservaron todos los botes llamados BigBoat.

```
SELECT s.sname
FROM sailors s
WHERE NOT EXISTS (
  SELECT 1 FROM boats b
  WHERE b.bname = 'BigBoat'
  AND NOT EXISTS (
    SELECT 1 FROM reserves r WHERE r.sid = s.sid AND r.bid = b.bid
  )
);
```

Respuesta:

$$\pi_{sname} ((\pi_{sid,bid} (reserves) \div \pi_{bid} (\sigma_{bname='BigBoat'} (boats))) \bowtie sailors)$$

8. Nombres de proveedores que suministran alguna parte roja.

```
SELECT DISTINCT s.sname
FROM suppliers s
JOIN catalog c ON c.sid = s.sid
JOIN parts p ON p.pid = c.pid
WHERE p.color = 'red';
```

Respuesta:

$$\pi_{sname} (suppliers \bowtie catalog \bowtie \sigma_{color='red'} (parts))$$

9. sid de proveedores que suministran alguna parte roja o verde.

```
SELECT DISTINCT c.sid
FROM catalog c JOIN parts p ON p.pid = c.pid
WHERE p.color IN ('red','green');
```

Respuesta:

$$\pi_{sid} (catalog \bowtie \sigma_{color='red' \vee color='green'} (parts))$$

10. sid de proveedores que suministran una parte roja o están en 21 George Street.

```
SELECT c.sid FROM catalog c JOIN parts p ON p.pid=c.pid WHERE p.color='red'
UNION
SELECT sid FROM suppliers WHERE address = '21 George Street';
```

Respuesta:

$$\pi_{sid}(catalog \bowtie \sigma_{color='red'}(parts)) \cup \pi_{sid}(\sigma_{address='21 George Street'}(suppliers))$$

11. sid de proveedores que suministran alguna parte roja y alguna verde.

```
SELECT sid FROM catalog c JOIN parts p ON p.pid=c.pid WHERE p.color='red'
INTERSECT
SELECT sid FROM catalog c JOIN parts p ON p.pid=c.pid WHERE p.color='green';
```

Respuesta:

$$\pi_{sid}(catalog \bowtie \sigma_{color='red'}(parts)) \cap \pi_{sid}(catalog \bowtie \sigma_{color='green'}(parts))$$

12. Pares de sid tales que el primero cobra más que el segundo por la misma parte.

```
SELECT c1.sid AS sid_caro, c2.sid AS sid_barato
FROM catalog c1
JOIN catalog c2 ON c1.pid = c2.pid AND c1.cost > c2.cost;
```

Respuesta:

$$\pi_{C1.sid, C2.sid}(\rho_{C1}(catalog) \bowtie_{C1.pid=C2.pid \wedge C1.cost > C2.cost} \rho_{C2}(catalog))$$

13. sid de proveedores que suministran solo partes rojas.

```
SELECT sid FROM suppliers
EXCEPT
SELECT c.sid FROM catalog c JOIN parts p ON p.pid=c.pid WHERE p.color <> 'red';
```

Respuesta:

$$\pi_{sid}(suppliers) - \pi_{sid}(catalog \bowtie \sigma_{color \neq 'red'}(parts))$$

14. sid de proveedores que suministran todas las partes.

```
SELECT c.sid
FROM catalog c
GROUP BY c.sid
HAVING COUNT(DISTINCT c.pid) = (SELECT COUNT(*) FROM parts);
```

Respuesta:

$$\pi_{sid, pid}(catalog) \div \pi_{pid}(parts)$$

15. Nombres y salarios de jefes que tienen algún empleado con salario > 100.

```
SELECT DISTINCT b.name, b.salary
FROM employees b
JOIN supervises sv ON sv.boss = b.number
JOIN employees e ON e.number = sv.employee
WHERE e.salary > 100;
```

Respuesta:

$$\pi_{B.name, B.salary} (\rho_B (employees) \bowtie_{B.number=sv.boss} (supervises \bowtie_{sv.employee=E.number} \rho_E (\sigma_{salary>100} (employees))))$$

16. Pares (jefe, empleado) donde el empleado gana más que su jefe.

```
SELECT b.name AS jefe, e.name AS empleado
FROM supervises sv
JOIN employees b ON b.number = sv.boss
JOIN employees e ON e.number = sv.employee
WHERE e.salary > b.salary;
```

Respuesta:

$$\pi_{B.name, E.name} (\sigma_{E.salary>B.salary} (\rho_B (employees) \bowtie_{B.number=sv.boss} supervises \bowtie_{sv.employee=E.number} \rho_E (employees)))$$

17. Nombres de empleados que no tienen jefe.

```
SELECT name FROM employees
WHERE number NOT IN (SELECT employee FROM supervises);
```

Respuesta:

$$\pi_{name} (employees) - \pi_{E.name} (\rho_E (employees) \bowtie_{E.number=supervises.employee} supervises)$$

18. Hora de cita y nombre del cliente para las citas de Giuliano el 2026-02-14.

```
SELECT c.name, a.ptime
FROM appointments a
JOIN staff s ON s.sid = a.sid
JOIN clients c ON c.cid = a.cid
WHERE a.adate = DATE '2026-02-14' AND s.name = 'Giuliano';
```

Respuesta:

$$\pi_{name, ptime} (clients \bowtie (\sigma_{adate='2026-02-14'} (appointments) \bowtie \sigma_{name='Giuliano'} (staff)))$$

19. Servicios que han sido solicitados al menos una vez.

```
SELECT DISTINCT service FROM appointments;
```

Respuesta:

$$\pi_{service} (appointments)$$

20. Clientes (nombre y teléfono) que nunca tomaron el servicio manicure.

```
SELECT c.name, c.phone
FROM clients c
WHERE c.cid NOT IN (SELECT cid FROM appointments WHERE service = 'manicure')
;
```

Respuesta:

$$\pi_{name, phone}(clients) - \pi_{name, phone}(clients \bowtie \sigma_{service='manicure'}(appointments))$$

2 De Álgebra Relacional a SQL

21. Expresión

$$\pi_{sid}(\sigma_{rating>7}(sailors))$$

Respuesta (SQL):

```
SELECT sid FROM sailors WHERE rating > 7;
```

22. Expresión

$$\pi_{sname}(\sigma_{age\geq 18 \wedge age\leq 25}(sailors))$$

Respuesta (SQL):

```
SELECT sname FROM sailors WHERE age >= 18 AND age <= 25;
```

23. Expresión

$$\pi_{sname}(sailors \bowtie reserves \bowtie \sigma_{color='red'}(boats))$$

Respuesta (SQL):

```
SELECT DISTINCT s.sname
FROM sailors s
JOIN reserves r ON r.sid = s.sid
JOIN boats b ON b.bid = r.bid
WHERE b.color = 'red';
```

24. Expresión

$$\pi_{S1.sid}(\rho_{S1}(sailors) \bowtie_{S1.rating>S2.rating} \rho_{S2}(\sigma_{sname='Bob'}(sailors)))$$

Respuesta (SQL):

```
SELECT s1.sid
FROM sailors s1, sailors s2
WHERE s2.sname = 'Bob' AND s1.rating > s2.rating;
```

25. Expresión

$$\pi_{sid}(sailors) - \pi_{S1.sid}(\rho_{S1}(sailors) \bowtie_{S1.rating<S2.rating} \rho_{S2}(sailors))$$

Respuesta (SQL):

```
SELECT sid FROM sailors
EXCEPT
SELECT s1.sid FROM sailors s1, sailors s2 WHERE s1.rating < s2.rating;
```

26. Expresión

$$\pi_{pname}(\sigma_{color='red'}(parts))$$

Respuesta (SQL):

```
SELECT pname FROM parts WHERE color = 'red';
```

27. Expresión

$$\pi_{cost} (\sigma_{color='red' \vee color='green'} (parts) \bowtie catalog)$$

Respuesta (SQL):

```
SELECT c.cost
FROM catalog c JOIN parts p ON p.pid = c.pid
WHERE p.color = 'red' OR p.color = 'green';
```

28. Expresión

$$\pi_{sid} (\sigma_{color='red' \vee color='green'} (parts) \bowtie catalog)$$

Respuesta (SQL):

```
SELECT DISTINCT c.sid
FROM catalog c JOIN parts p ON p.pid = c.pid
WHERE p.color IN ('red','green');
```

29. Expresión

$$\pi_{sname} (\pi_{sid} (\sigma_{color='red' \vee color='green'} (parts) \bowtie catalog) \bowtie suppliers)$$

Respuesta (SQL):

```
SELECT DISTINCT s.sname
FROM suppliers s
JOIN catalog c ON c.sid = s.sid
JOIN parts p ON p.pid = c.pid
WHERE p.color IN ('red','green');
```

30. Expresión

$$\pi_{sname} (\sigma_{color='red'} (parts) \bowtie \sigma_{cost<100} (catalog) \bowtie suppliers)$$

Respuesta (SQL):

```
SELECT DISTINCT s.sname
FROM suppliers s
JOIN catalog c ON c.sid = s.sid
JOIN parts p ON p.pid = c.pid
WHERE p.color = 'red' AND c.cost < 100;
```

31. Expresión

$$\pi_{sname} (\sigma_{color='red'} (parts) \bowtie \sigma_{cost<100} (catalog) \bowtie suppliers) \cap \pi_{sname} (\sigma_{color='green'} (parts) \bowtie \sigma_{cost<100} (catalog) \bowtie suppliers)$$

Respuesta (SQL):

```
SELECT s.sname FROM suppliers s JOIN catalog c ON c.sid=s.sid JOIN parts p
    ON p.pid=c.pid
WHERE p.color='red' AND c.cost<100
INTERSECT
SELECT s.sname FROM suppliers s JOIN catalog c ON c.sid=s.sid JOIN parts p
    ON p.pid=c.pid
WHERE p.color='green' AND c.cost<100;
```

32. Expresión

$$\pi_{sid}(\sigma_{color='red'}(parts) \bowtie \sigma_{cost < 100}(catalog)) \cap \pi_{sid}(\sigma_{color='green'}(parts) \bowtie \sigma_{cost < 100}(catalog))$$

Respuesta (SQL):

```
SELECT c.sid FROM catalog c JOIN parts p ON p.pid=c.pid WHERE p.color='red'
      AND c.cost<100
INTERSECT
SELECT c.sid FROM catalog c JOIN parts p ON p.pid=c.pid WHERE p.color='green'
      AND c.cost<100;
```

33. Expresión

$$\pi_{sid}(suppliers) - \pi_{sid}(catalog)$$

Respuesta (SQL):

```
SELECT sid FROM suppliers
EXCEPT
SELECT sid FROM catalog;
```

34. Expresión

$$(\pi_{sid}(catalog) \times \pi_{pid}(parts)) - \pi_{sid,pid}(catalog)$$

Respuesta (SQL):

```
SELECT c.sid, p.pid
FROM (SELECT DISTINCT sid FROM catalog) c, parts p
EXCEPT
SELECT sid, pid FROM catalog;
```

35. Expresión

$$\pi_{name}(\sigma_{salary > 1000 \wedge age < 30}(employees))$$

Respuesta (SQL):

```
SELECT name FROM employees WHERE salary > 1000 AND age < 30;
```

36. Expresión

$$\pi_{B.name}(\rho_B(employees) \bowtie_{B.number=sv.boss} \rho_{sv}(supervises))$$

Respuesta (SQL):

```
SELECT DISTINCT b.name
FROM employees b
JOIN supervises sv ON sv.boss = b.number;
```

37. Expresión

$$\pi_{number}(employees) - \pi_{boss}(supervises)$$

Respuesta (SQL):

```
SELECT number FROM employees
EXCEPT
SELECT boss FROM supervises;
```

38. Expresión

$$\pi_{name, phone}(clients \bowtie \sigma_{service='haircut'}(appointments))$$

Respuesta (SQL):

```
SELECT DISTINCT c.name, c.phone
FROM clients c
JOIN appointments a ON a.cid = c.cid
WHERE a.service = 'haircut';
```

39. Expresión

$$\pi_{cid}(clients) - \pi_{cid}(appointments)$$

Respuesta (SQL):

```
SELECT cid FROM clients
EXCEPT
SELECT cid FROM appointments;
```

40. Expresión

$$\pi_{C.name, S.name}(\rho_C(clients) \bowtie_{C.cid=a.cid} \rho_a(appointments) \bowtie_{a.sid=S.sid} \rho_S(staff))$$

Respuesta (SQL):

```
SELECT DISTINCT c.name, s.name
FROM clients c
JOIN appointments a ON a.cid = c.cid
JOIN staff s ON s.sid = a.sid;
```